INCH-POUND
MIL-DTL-16377/23B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/23A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 200 WATTS, 115 VOLTS, WATERTIGHT, SYMBOLS 57, 57.1, 69, 69.1, 103, 57SSL, 57.1SSL, 69SSL, 69.1SSL, AND 103SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 16 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

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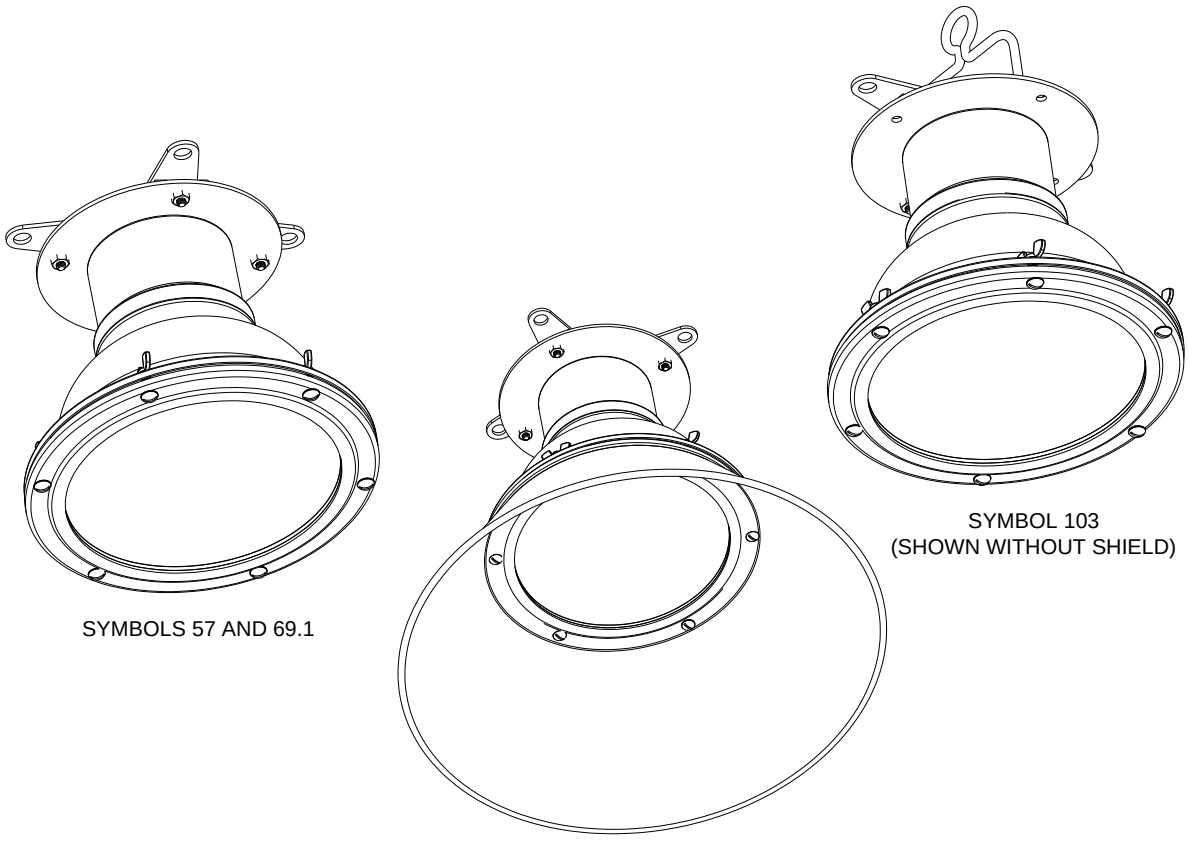


FIGURE 1. Typical view of symbols 57, 57.1, 69, 69.1, and 103.

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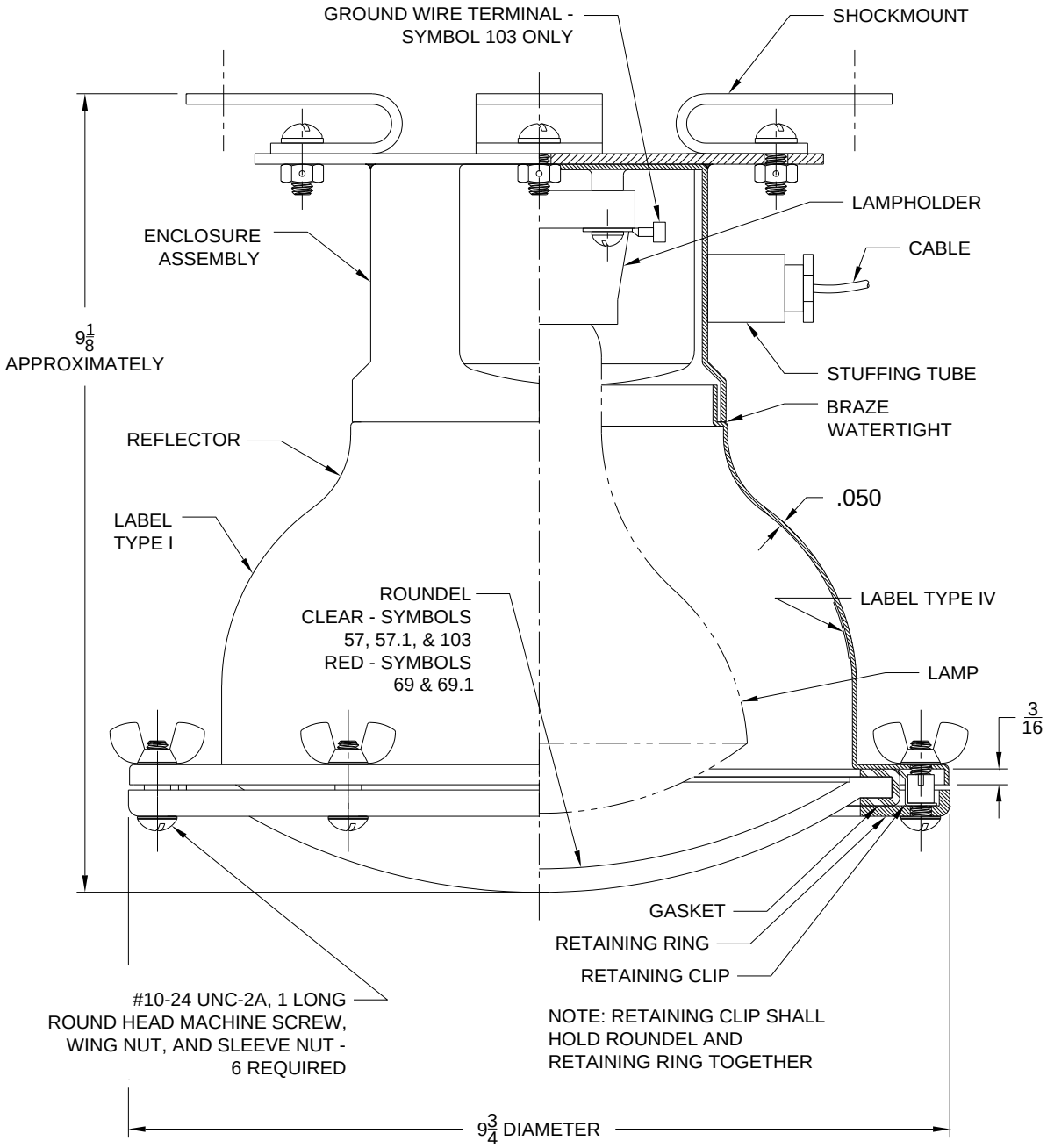


FIGURE 2. Fixture assembly, symbols 57 and 69.1.

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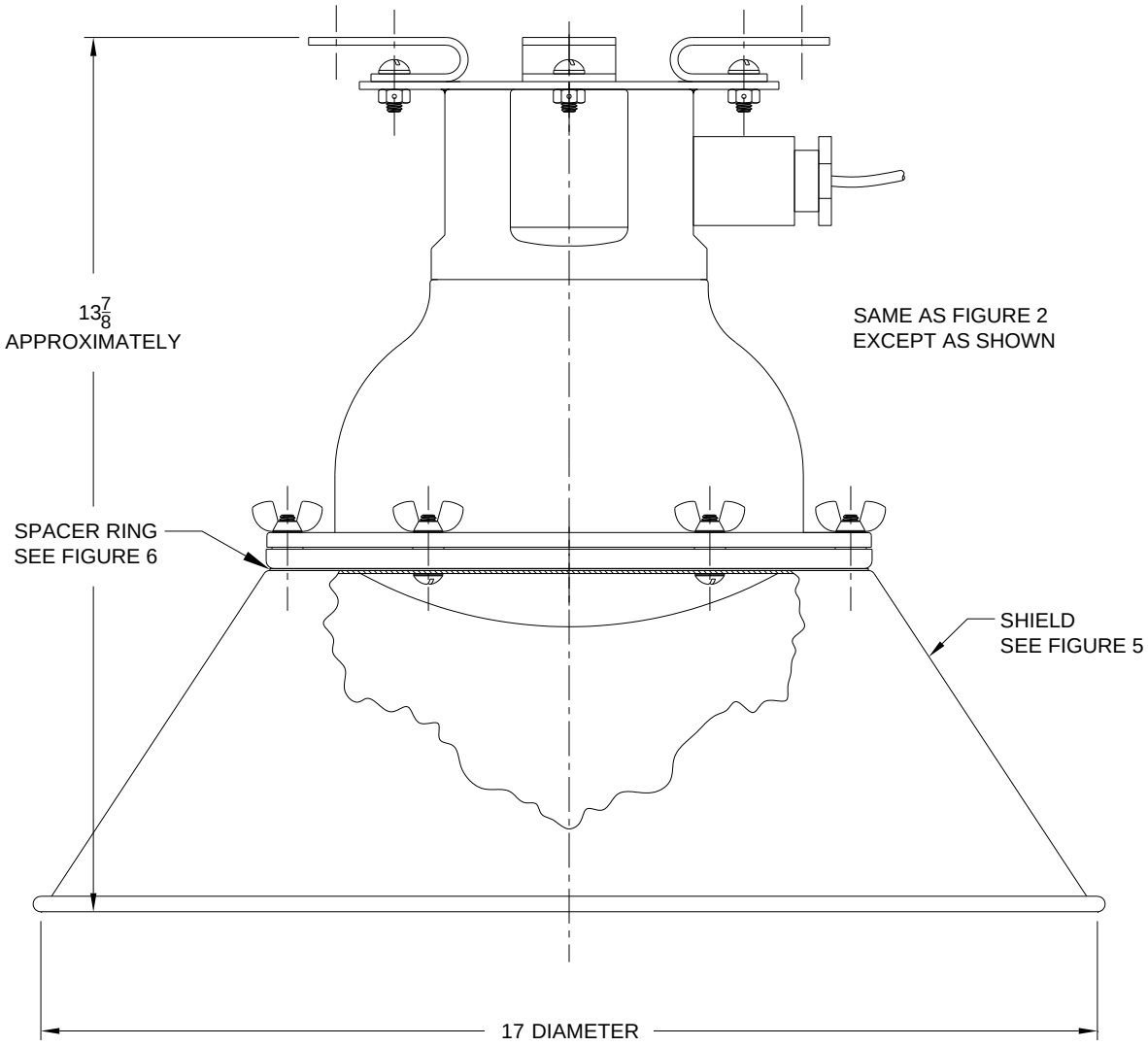
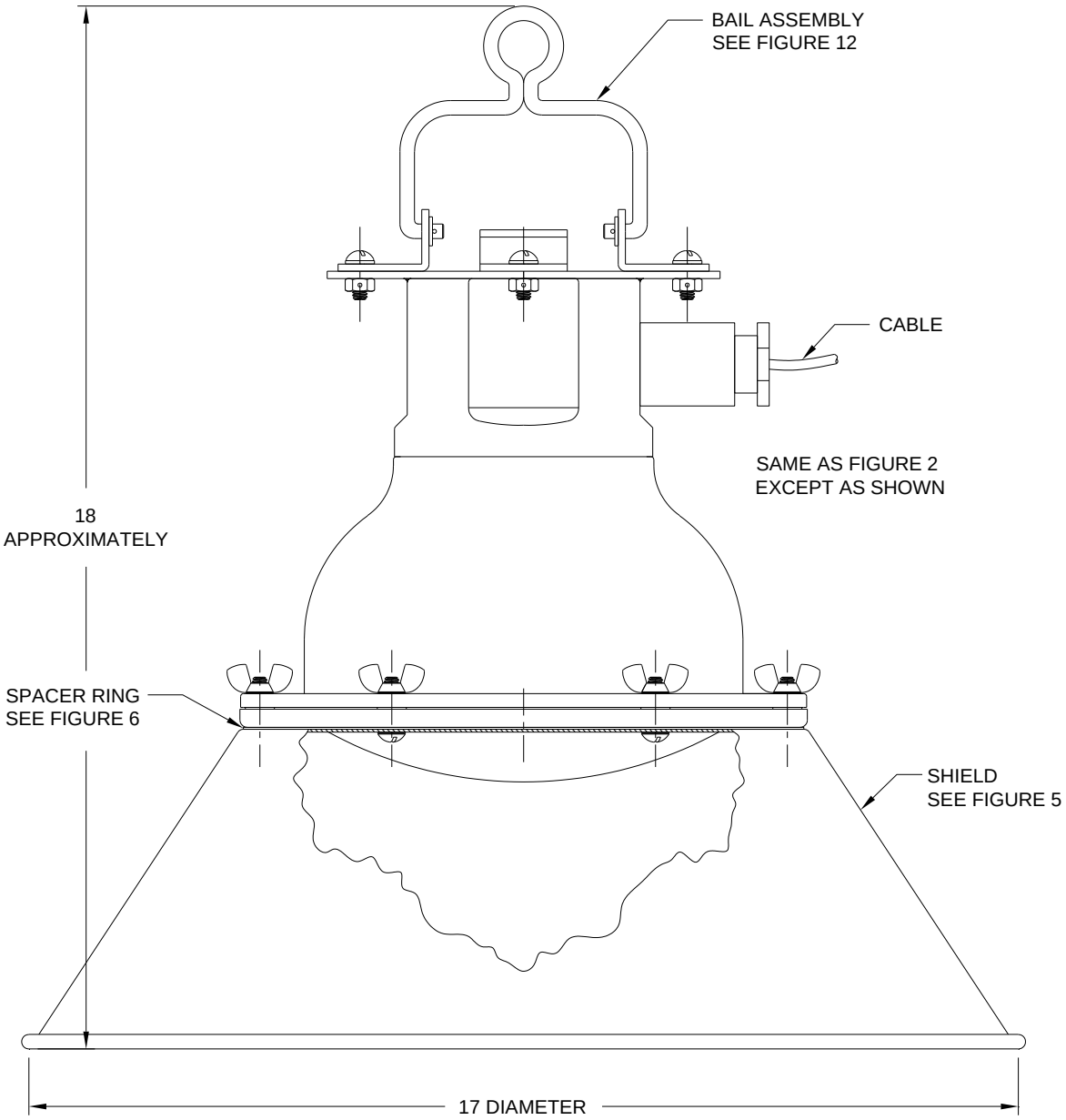


FIGURE 3. Fixture assembly, symbols 57.1 and 69.

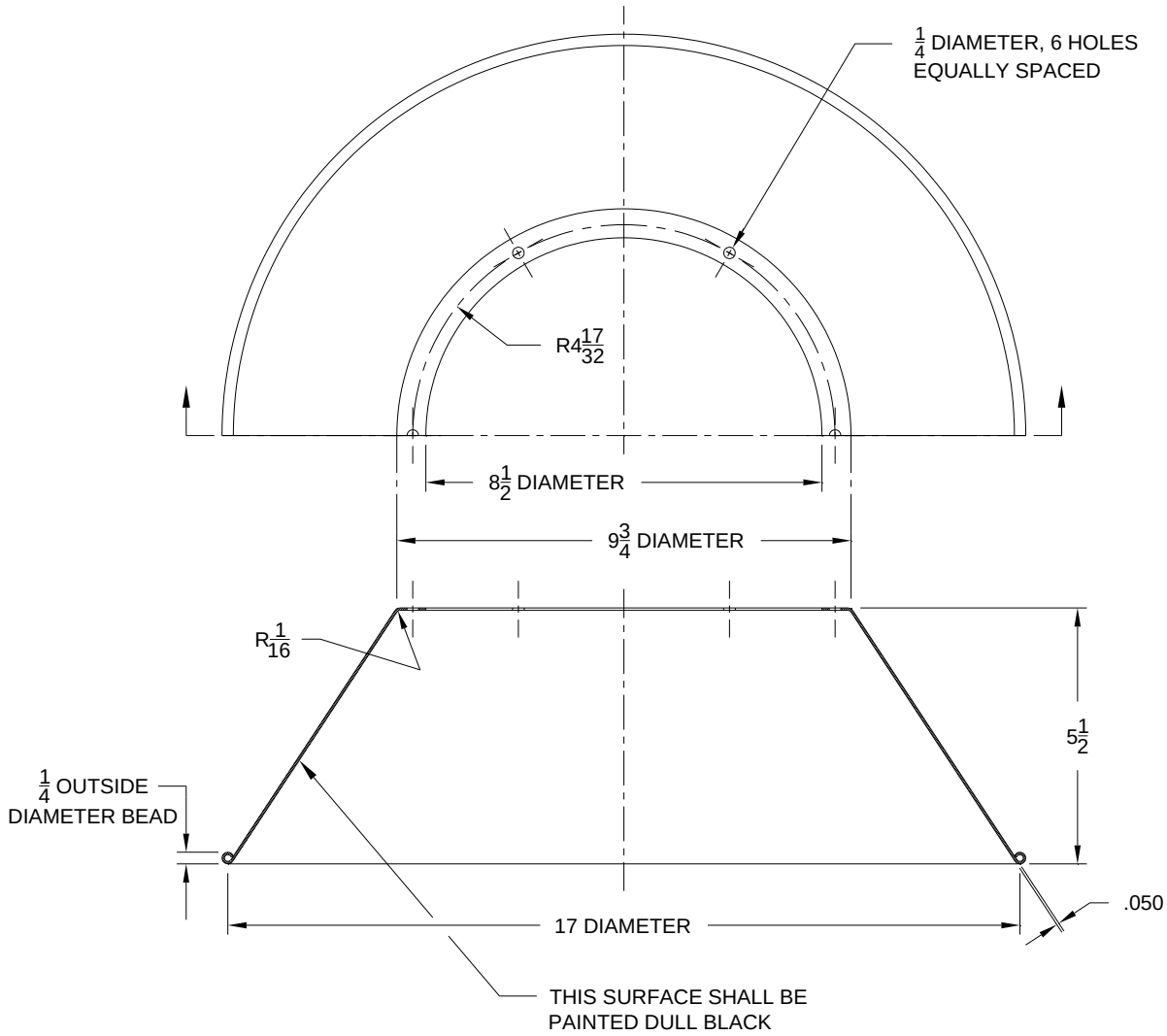
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NOTE: See figure 2 for ground wire attachment.

FIGURE 4. Fixture assembly, symbol 103.

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NOTE: See 3.8.4 of MIL-DTL-16377 for definition of dull black.

FIGURE 5. Shield.

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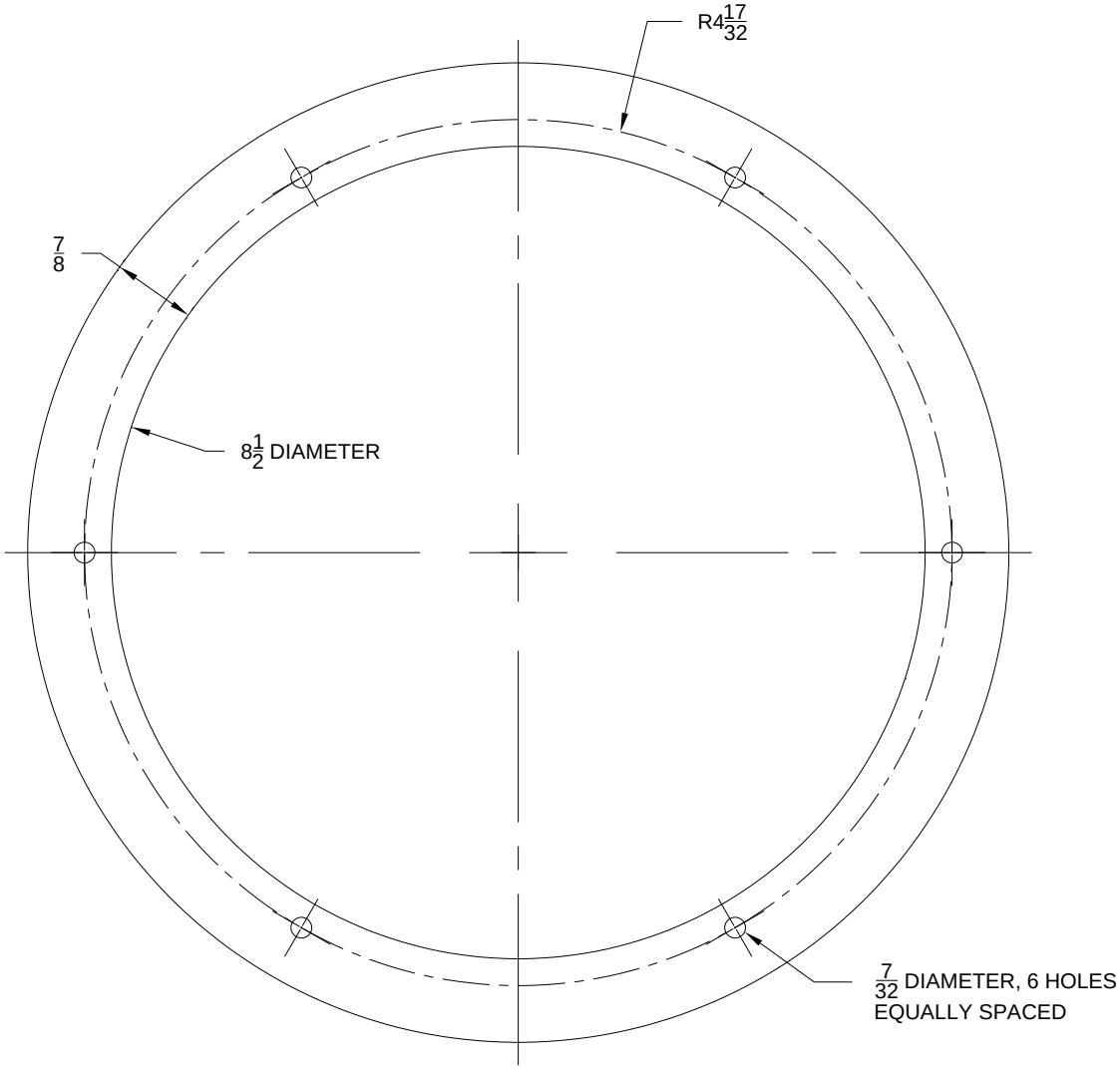


FIGURE 6. Spacer ring.

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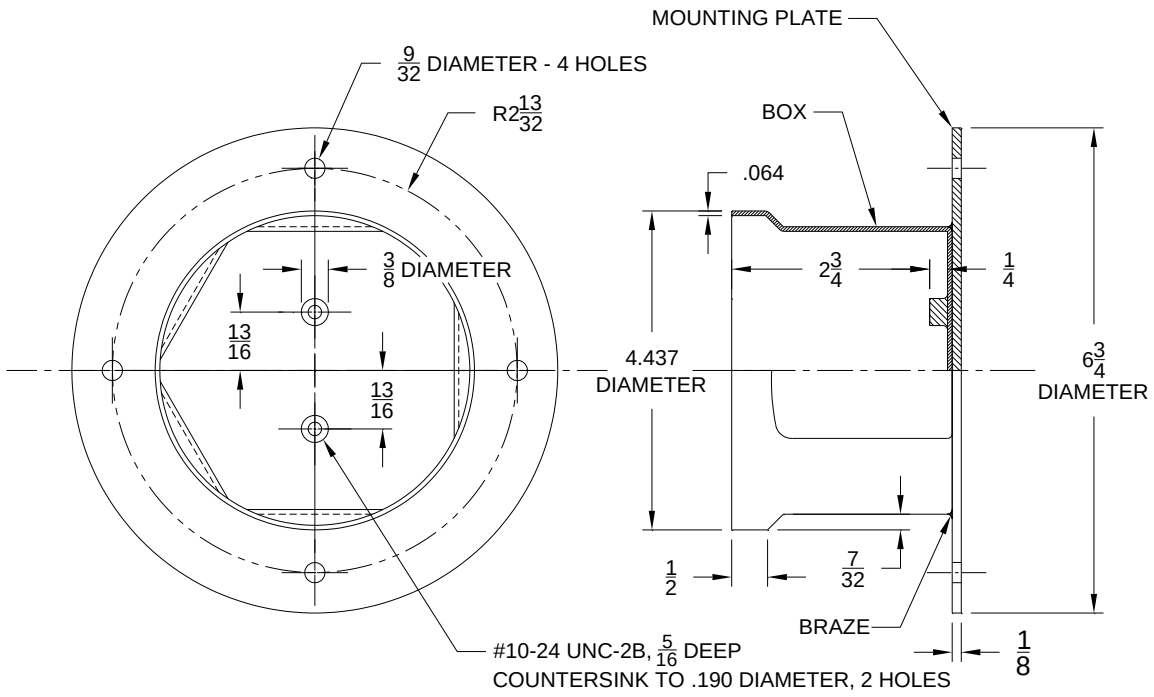


FIGURE 7. Enclosure.

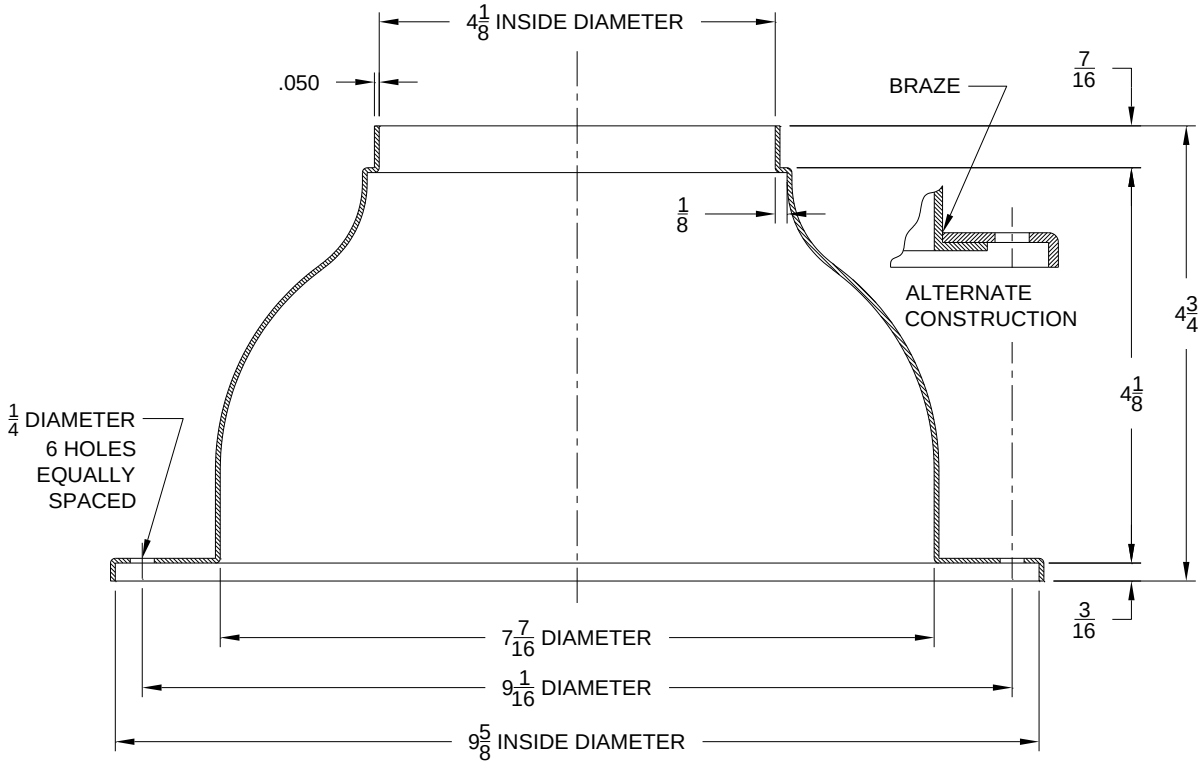


FIGURE 8. Reflector.

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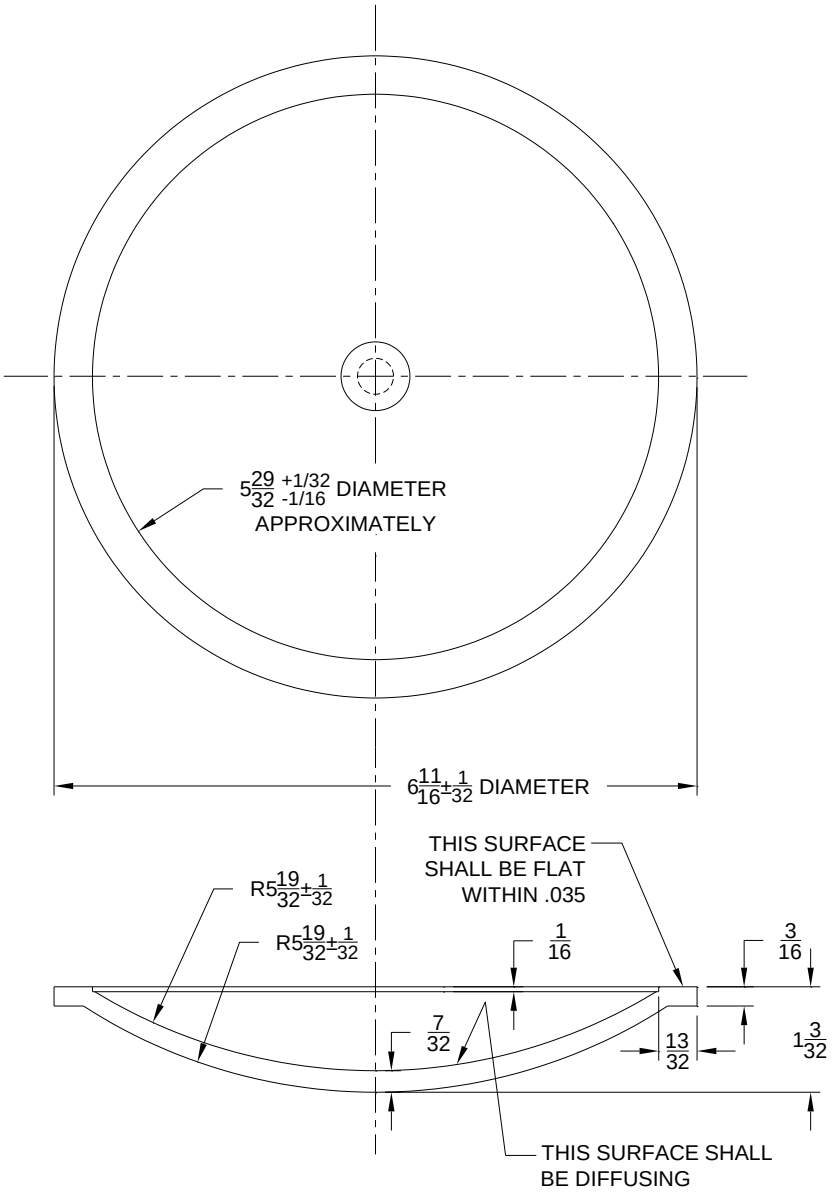


FIGURE 9. Roundel.

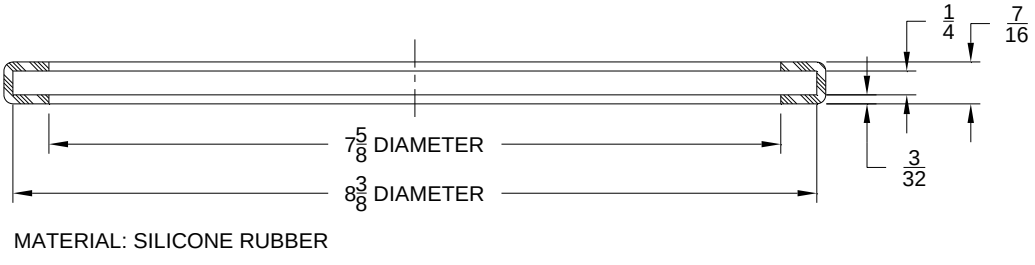


FIGURE 10. Gasket.

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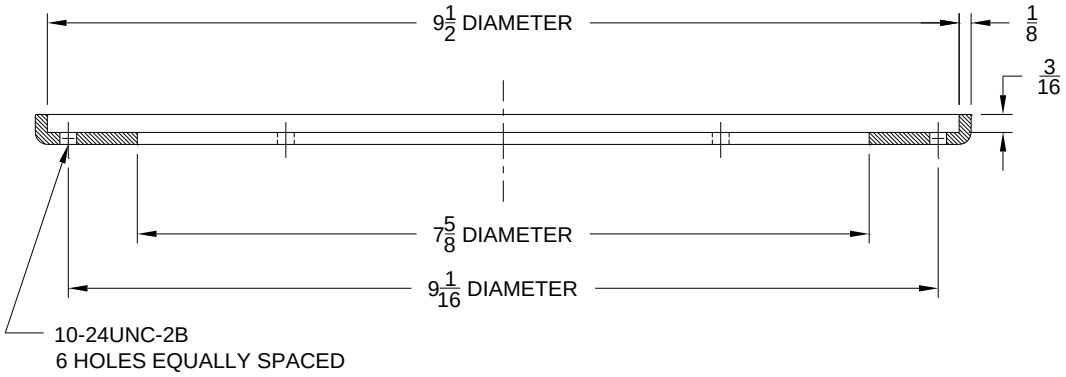


FIGURE 11. Retaining ring.

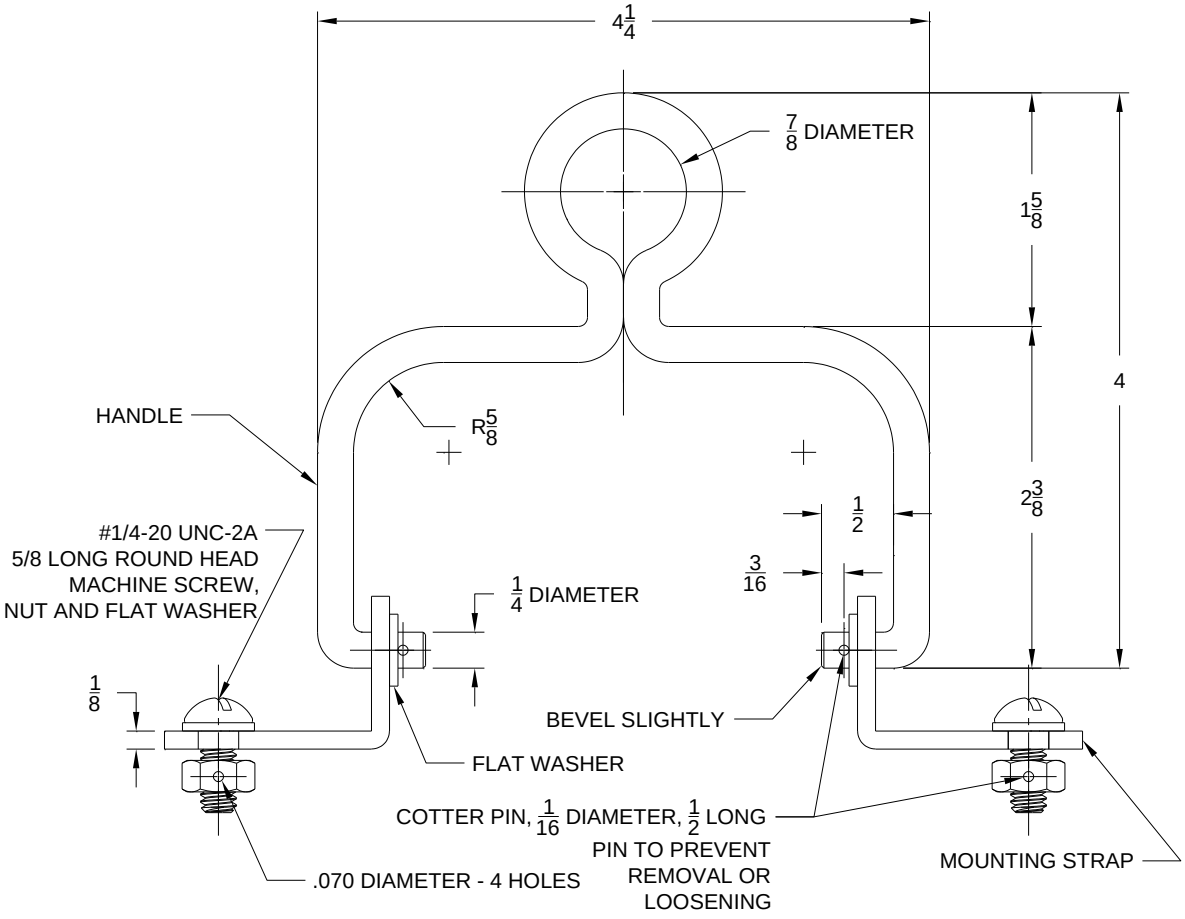


FIGURE 12. Bail assembly.

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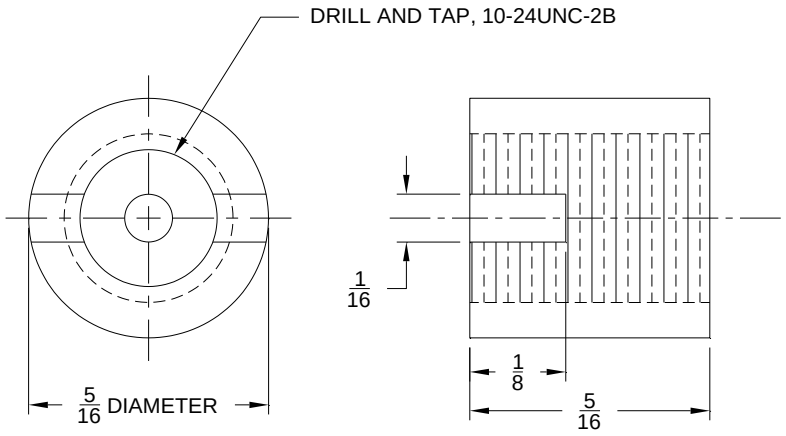


FIGURE 13. Sleeve nut.

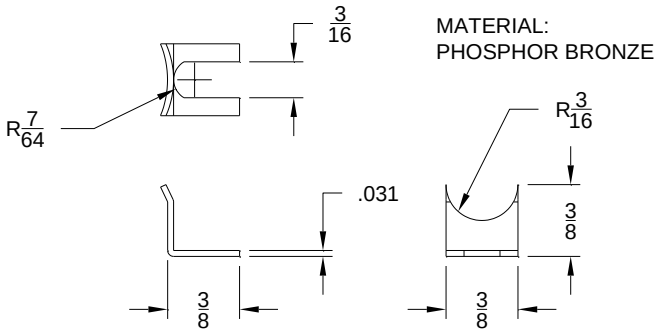


FIGURE 14. Lens retaining ring.

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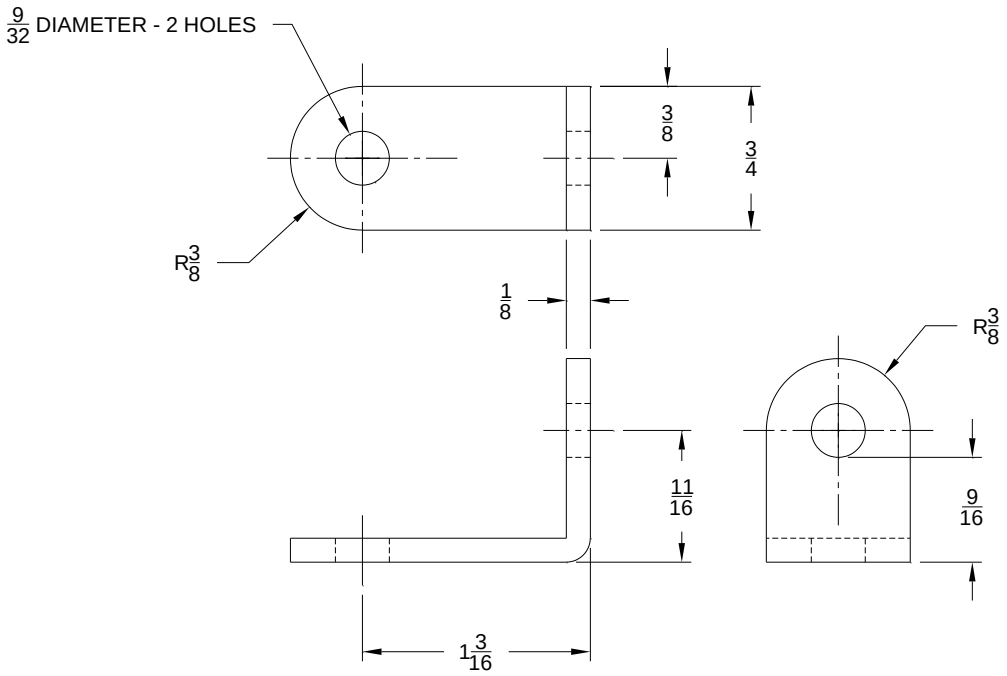
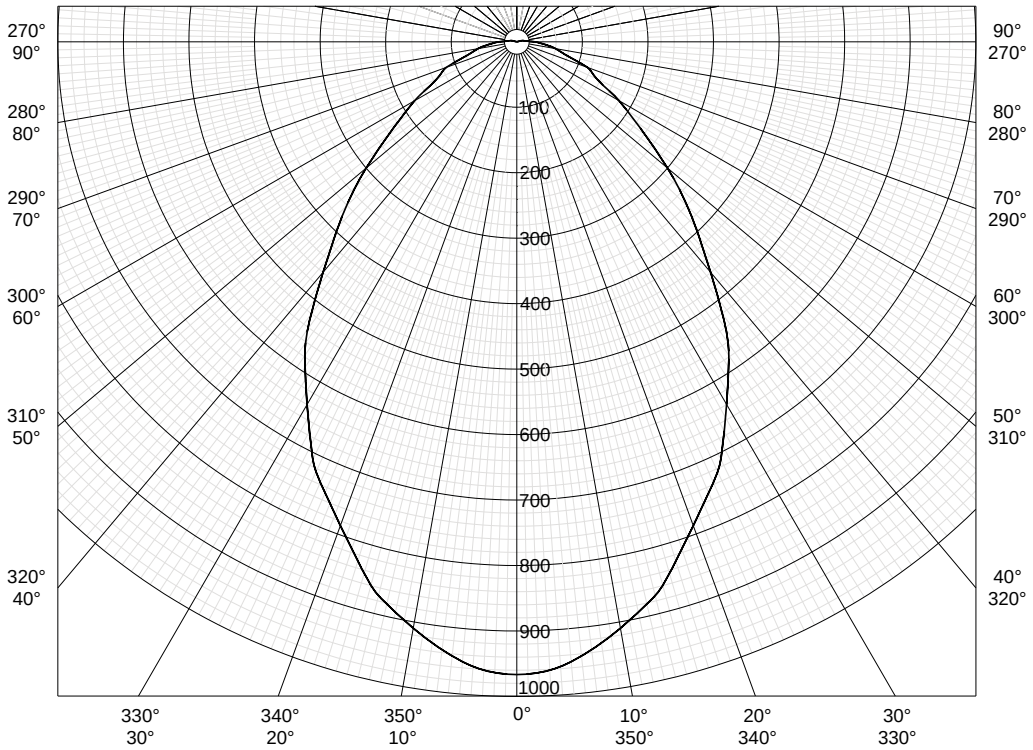


FIGURE 15. Mounting strap.

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ANGLE	CANDLEPOWER	
	SYM 57, 103	SYM 69
0	967	---
5	951	---
10	930	---
15	860	---
20	787	---
25	725	---
30	640	---
35	562	---
40	460	---
45	377	---
50	300	---
55	230	---
60	180	---

ANGLE	CANDLEPOWER	
	SYM 57, 103	SYM 69
65	135	---
70	115	---
75	77	---
80	60	---
85	42	---
90	27	---
95	14	---
100	10	---
105	6	---
110	0	---
115	---	---
120	---	---
125	---	---

Photometric test data:

Distance: 10 feet

Voltage: 115 volts

Ambient temperature: 76 °F

Lamp: 200 watts, rough service conforming to a lamp in accordance with incandescent requirements section, furnished by laboratory for test, shall be used

FIGURE 16. Candlepower distribution curves.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

57 and 57.1: Fixture, clear transparent diffusing roundel, permanently mounted, for use with 200-watt incandescent lamp.
69 and 69.1: Fixture, red transparent roundel, permanently mounted, for use with 200-watt incandescent lamp.
103: Fixture, clear transparent diffusing roundel, portable, with bracket, for use with 200-watt incandescent lamp.

DESIGN

Electrical rating:	200 watts, 115 volts.
Dimensions and configuration:	See figures 1 through 15.
Mounting dimensions:	See figure 28 of MIL-DTL-16377.
Photometric data:	See figure 16 (symbols 57, 57.1, and 103 only).
Material:	Unless otherwise specified, material shall be brass.
Roundels:	
Symbols 57, 57.1, and 103:	Clear transparent diffusing; see figure 9.
Symbols 69 and 69.1:	Red transparent, part number MS24489-1 of MS24489.
Lampholder:	UL 496, Supplement SB, double-contact, medium-screw.
Shock mounts:	MIL-F-16377/41, part number M16377/41-029.
Enclosure:	Watertight.
Stuffing tube:	Shall be in accordance with MIL-S-24235.
Stuffing tube packing:	Shall be symbol 2447.2 of S6202-1197101, except that material shall be silicone rubber, capable of withstanding constant temperature inside the fixture of 338 °F (170 °C).
Weight:	
Symbols 57 and 69.1:	9 pounds, 0 ounces maximum.
Symbols 57.1 and 69:	11 pounds, 8 ounces maximum.
Symbol 103:	12 pounds, 8 ounces maximum.
Lamp:	These fixtures shall be designed to accommodate a 200-watt, 115-volt, R-40, reflector flood, medium-screw base, rough service lamp conforming to industry number 200RL/RS/115V. (Lamp not furnished.)
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2: (a) One type I. (b) One type IV. Apply one type IV label with the following information in 3/16-inch letters: "USE THE FOLLOWING REPAIR PARTS ONLY: LAMP NSN 6420-00-248-4280, WHITE LENS NSN 6210-00-299-2893, RED LENS NSN 6210-00-633-6886."

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SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS:

57SSL and 57.1SSL: Fixture, clear transparent diffusing, permanently mounted, solid state.
69SSL and 69.1SSL: Fixture, red transparent, permanently mounted, solid state.
103SSL: Fixture, clear transparent diffusing, portable, with bracket, solid state.

DESIGN

Electrical rating:	100 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 4.
Photometric data:	See figure 16 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 57SSL and 57.1SSL:	See figure 2 and figure 28 of MIL-DTL-16377.
Symbols 69SSL and 69.1SSL:	See figure 3 and figure 28 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Weight:	
Symbols 57SSL and 69.1SSL:	9 pounds, 0 ounces maximum.
Symbols 57.1SSL and 69SSL:	11 pounds, 8 ounces maximum.
Symbol 103SSL:	12 pounds, 8 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type I and table III for type III.
Repair parts:	See table II.
Conformance:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.

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TABLE I. First article and comparison inspection for complete type II fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	Symbols 57, 57.1, 69, and 69.1
Vibration	3.5.13	4.8.8	Symbols 57, 57.1, 69, and 69.1
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Grounding circuit	3.6.13.3	4.8.11	Symbol 103 only
Continuity of grounding	3.6.13.4	4.8.12	Symbols 57, 57.1, 69, and 69.1
Leakage current	3.6.13.5	4.8.13	Symbol 103 only
Optical uniformity	3.7.4.1.1	4.8.4	---
Mechanical shock	3.7.5.6	4.8.17.3	6 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3.7	4.8.6.3	Symbols 57, 57.1, and 103
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbols 69 and 69.1
Light output	3.7.1	4.8.6.1	Symbols 57, 57.1, and 103
Magnetic permeability	3.4.8	4.8.18	---
Salt fog	3.3.15	4.8.10	---

TABLE II. First article and comparison inspection for repair parts.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.4.1.1	4.8.4	1/
Mechanical shock	3.7.5.6	4.8.17.3	6 foot-pounds 1/
Thermal shock	3.7.5.7	4.8.17.4	1/
Breakage pattern	3.7.5.9	4.8.17.6	1/
Color and luminous transmittance	3.7.3.1	4.8.6.3	Red 1/
Color and luminous transmittance	3.7.3.7	4.8.6.3	Clear 1/

NOTE:
1/ Roundels only.

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TABLE III. First article and comparison inspection for complete type III fixtures. [1/]

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Grounding circuit	3.6.13.3	4.8.11	Symbol 103SSL only
Leakage current	3.6.13.5	4.8.13	Symbol 103SSL only
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	[1/]
Insulation resistance	3.6.16	4.8.3	[1/]
Enclosure effectiveness	3.5.11	4.8.14.4	[1/] Watertight
Shock	3.5.12	4.8.7	[1/]
Vibration	3.5.13	4.8.8	[1/]
Enclosure effectiveness	3.5.11	4.8.14.4	[1/] Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[1/]
Insulation resistance	3.6.16	4.8.3	[1/]
Noise	3.5.14	4.8.9	[1/]
Continuity of grounding	3.6.13.4	4.8.12	[1/]
Photometrics	3.7.1	4.8.6	---
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.6.17	4.8.22	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTES:

[1/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. Part numbers and general information.

Military Part [1/] Number M16377/23-	Item Name	National Stock Number (NSN)	Remarks
57	Complete fixture, symbol 57	6210-00-283-9691	---
57.1	Complete fixture, symbol 57.1	---	[2/]
69	Complete fixture, symbol 69	---	[3/]
69.1	Complete fixture, symbol 69.1	---	[4/]
103	Complete fixture, symbol 103	---	---
57SSL	Complete fixture, solid state, symbol 57SSL	---	---
57.1SSL	Complete fixture, solid state, symbol 57.1SSL	---	---
69SSL	Complete fixture, solid state, symbol 69SSL	---	---
69.1SSL	Complete fixture, solid state, symbol 69.1SSL	---	---
103SSL	Complete fixture, solid state, symbol 103SSL	---	---
001	Roundel, clear transparent	6210-00-299-2893	---
002	Roundel, red transparent	---	[5/]
003	Bail assembly	6210-00-399-7120	---
004	Shield and spacer assembly	6210-00-548-0107	Figures 5 and 6
005	Gasket	---	Figure 10

NOTES:

1/ Formerly covered on NAVSHIPS drawing 9000-6401-74220 and MS17276.

2/ Symbol 57.1 is made up of parts M16377/23-57 and MS16377/23-004.

3/ Symbol 69 is made up of parts M16377/23-57 (less M16377/23-001) and M16377/23-002.

4/ Symbol 69.1 is made up of parts M16377/23-69 and M16377/23-004.

5/ NSN 6210-00-548-0228 for M16377/23-002 has been superseded by MS24489-1.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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